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Robust audio copy-move forgery detection on short forged slices using sliding window

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ABSTRACT

Audio copy-move forgery detection (CMFD), which aims at finding possible forgeries that are derived from the same audio recording, has been one of the most difficult challenges in blind audio forensics. However, the existing works rarely pay attention to the forgery of short slices within voiced segments, which may result in missed detection. In this work, we present a robust audio CMFD method on the basis of sliding window (SW) and constant Q cepstral coefficients (CQCC). Specifically, we first propose two SW strategies to illustrate how to reveal the possible copy-move forgeries within and between voiced segments. Then, we combine the CQCC feature with the Pearson correlation coefficients to measure the similarity between different short slices. Finally, we evaluate the proposed method, named SW-CQCC, against the state-of-the-art approaches to CMFD on real-world read English and Chinese corpora. Experimental results demonstrate that our SW-CQCC exhibits significantly high effectiveness and robustness in detecting short forged slices within voiced segments, which may be helpful to audio forensic examinations.

1. Introduction

With the development of the digital audio technology, audio recordings used as evidence have become increasingly important to litigation. However, before their admissibility as evidence, they must be able to demonstrate that no changes, deletions, additions or other alterations have occurred. Accordingly, an audio forensic expert is often required to help determine whether the submitted audio recordings are tampered.

Usually, audio tampering includes inserting, deleting, copying, pasting, splicing, and synthesized forgeries [1,2]. The main purpose of tampering is to destroy, distort, or forge new semantics to quote out of the context and intentionally gloss over details. In this regard, the copy-move forgery refers to copying and pasting some voiced segments to other locations within the same audio recording and can be easily achieved by various powerful audio editors. For example, the sentence “I’m guilty” can be tampered to “I’m not guilty” by copying and inserting the word “not” said by the same speaker within the same recording. Since the forged segments are highly consistent with the original recording in many basic characteristics (e.g., amplitude, frequency, tone, and speed), it is difficult for audio forensic experts to detect and localize copy-move forgeries, especially very short forged clips. Therefore, the audio copy-move forgery detection (CMFD) has become one of the most challenging topics in the field of blind audio forensics [3].

Nowadays, a large amount of research work has been done in image CMFD [4–6]. Most of the existing audio CMFD methods are derived from the ideas of these image-based CMFD [3]. Specifically, voiced segments are first distinguish from the audio recording by the audio segmentation techniques, such as voice activity detection (VAD) [7,8] or “YAAPT” [9]. Next, the acoustic feature of each voiced segment is extracted for the similarity evaluation between each pair of voiced segments. Although the existing methods were empirically validated and technically feasible, it should be noted that these methods rely heavily on audio segmentation techniques to delimit voiced segments with clear boundaries among each other, and mainly focus on the similarity between long voiced segments. Intuitively, Fig. 1 depicts different tampering ways of audio copy-move forgeries. Most of the existing works pay attention to the forgery of the entire voiced segment (i.e., Type 1), but does not consider the possible forgery of short slices derived from voiced segments (i.e., Types 2–4), especially when the source and the target slices are within the same voiced segment (i.e., Type 4). This is because audio segmentation techniques can only

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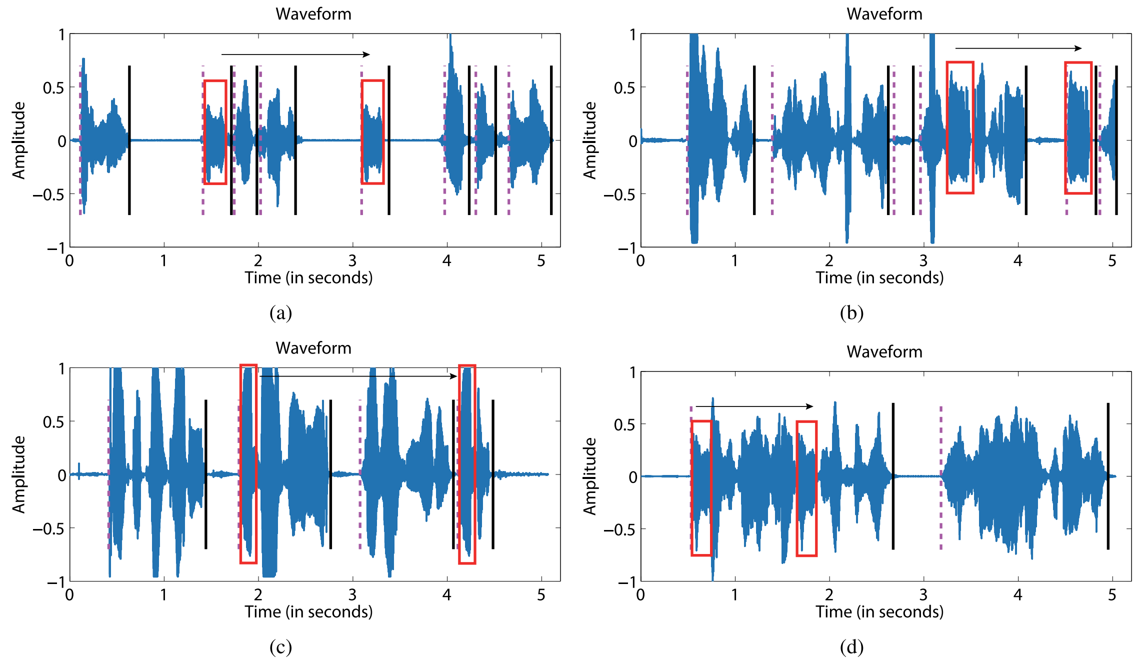

Fig. 1. Illustration of different tampering ways of audio copy-move forgeries. (a). Type 1: a voiced segment is completely forged as another independent voiced segment. (b). Type 2: a short slice of a voiced segment is forged as an independent voiced segment. (c). Type 3: a short slice of a voiced segment is forged as a slice of another voiced segment. (d). Type 4: a short slice of a voiced segment is forged as another slice within the same voiced segment. Note that the vertical dotted line and solid line indicate the initiation position and the termination position of a voiced segment divided by VAD, respectively.

find out long voiced segments that may contain very short forged slices. When comparing these long voiced segments through similarity, there is a high probability that these voiced segments will be judged to be similar, thus missing the detection of the short forged slices embedded in the long voiced segments. In addition, the shorter the forged slices are, the more difficult it is to detect these slices. For instance, a common gun shot lasts for less than 200 ms, and thus it is very difficult to identify which gun shots may be forged from the submitted audio recording of the case scene.

Against this background, this research focuses on CMFD with respect to different tampering ways of copy-move forgeries shown in Fig. 1 and makes the following contributions to the state-of-the-art in audio CMFD.

(1) We use constant Q cepstral coefficients (CQCC) [10–12] and PCCs to compute the similarity between different short, voiced slices.

(2) We propose two sliding window (SW) strategies to illustrate how to reveal all the suspicious copy-move forgeries within and between voiced segments.

(3) We evaluate our method, named SW-CQCC, against the state-of-the-art CMFD approaches on real-world read English and Chinese corpora. Encouragingly, experimental results demonstrate the high competitiveness of SW-CQCC on finding short forged slices, especially within the same voiced segment, which further improves the robustness of the audio CMFD.

The remainder of this work is organized as follows. First, in Section 2, an overview of the related work is provided. Section 3 introduces the CQCC feature extraction. Section 4 presents the method of similarity computation. Section 5 shows the details of the proposed CMFD method. Section 6 gives the experimental settings, while Section 7 illustrates experimental results. Last, Section 8 concludes this work.

2. Related work

To our best knowledge, Xiao et al. [13] made the first attempt to use the fast convolution algorithm to calculate the similarity of waveform between voiced segments. Yan et al. [14] extracted the pitch of every syllable and calculated the similarity between pitch sequences (PS) on the basis of Pearson correlation coefficients (PCCs) [23] and the average difference (AD). In their serial work, they [3] proposed a robust CMFD method, named PFS-DTW, in which the pitch tracking method "YAAPT" [9] is used to partition the audio signal into very short voiced segments, both pitch and formant sequences (PFS) are extracted as the feature set of a voiced segment, and the dynamic time warping (DTW) distance is applied to compute the similarity between different feature sets. However, when the pasted segment comes from continuous segments with strong correlation, PFS-DTW may not be able to accurately distinguish the voiced segments, so that the missed detection rate may increase. Additionally, when the duration of the pasted segment is short, PFS-DTW can only extract very short pitch sequences, which may lead to the increase of the error detection rate.

To reduce computational complexity, Li et al. [15] applied the filter to compare the difference of PS. Huang et al. [16] and Liu and Lu [17] extracted the feature of each voiced segment by discrete Fourier transform (DFT) and used PCCs to measure the similarity. Wang et al. [18] adopted discrete cosine transform (DCT) and singular value decomposition to extract the features and used the Euclidean distance to compute the similarity. Imran et al. [19] and Kucukugurlu et al. [20] applied a 1D local binary pattern operator (LBP) to calculate histograms and used mean square error (MSE) to compute the similarities of histograms. Xie et al. [21] first extracted four features by gammatone, Mel-frequency cepstral coefficients (MFCC) [24], PS, and DFT, respectively, and used PCCs or AD to evaluate the similarity on each feature. Then, they adopted C4.5 decision tree to merge the detection result of four features. Ustubioglu et al. [22] divided the speech into voiced and unvoiced segments by extracting pitch sequences from the speech signal. Then, they extracted the modified discrete cosine transform (MDCT) coefficients from the voiced segments, used the transposed mean of the coefficient matrix as the main acoustic feature, and introduced Euclidean distance (ED) to calculate the similarity between features.

Although the above methods were empirically validated and technically feasible, it should be noted that these methods rely heavily on audio segmentation techniques (e.g., VAD and YAAPT) to delimit voiced segments with clear boundaries among each other, and mainly

Table 1
Summary of research differences between the existing work and this study.

Authors	Audio segmentation technique	Targeted forgery type
Xiao et al. [13]	Fixed length	Type 1
Yan et al. [14]	YAAPT	Type 1
Li et al. [15]	YAAPT	Type 1
Huang et al. [16], Liu and Lu [17]	VAD	Type 1
Wang et al. [18]	VAD	Type 1
Imran et al. [19], Kucukugurlu et al. [20]	YAAPT	Type 1
Xie et al. [21]	VAD	Type 1
Ustubioglu et al. [22]	YAAPT	Type 1
Yan et al. [3]	YAAPT	Type 1
This study	VAD and SW	Types 1–4

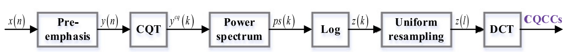

Fig. 2. Block diagram of the CQCC feature extraction.

focus on the similarity between long voiced segments (i.e., Type 1), as shown in Table 1. For Types 2–4, the division of VAD and YAAPT may not be always effective, which has a negative impact on the subsequent similarity comparison. Differently, this study combines VAD and SW strategies to uncover all the possible short forgeries (i.e., Types 1–4) within and between voiced segments.

3. CQCC feature set

CQCC is a newly popular feature for automatic speaker verification [10–12] on the basis of the constant Q transform (CQT) [25–27]. CQT transforms the time-domain signal into the time–frequency domain, so that a higher frequency resolution at low frequencies and a higher time resolution at high frequencies can be achieved. Specifically, for low-frequency waves, the bandwidth of CQT is quite small, but it has higher frequency resolution to decompose similar notes. On the other hand, for high-frequency waves, the bandwidth of CQT is relatively large, and it has higher time resolution to track rapidly changing overtones at high frequencies. Therefore, CQCC is able to capture greater spectral details at lower frequencies and greater temporal details at higher frequencies, which is quite different from the traditional time–frequency analysis methods [11].

In this work, we extract CQCC as the main acoustic features. As shown in Fig. 2, the extraction process of CQCC can be summarized as below [10].

(1) To smooth the spectrum and reduce noise, a pre-emphasis filter (e.g., high-pass filter) is applied to the audio signal $x(n)$ to amplify the high frequencies:

$$y(n) = x(n) - \alpha x(n-1) \tag{1}$$

where n denotes the sampling point index, and α is the filter coefficient.

(2) Perform CQT on $y(n)$ to obtain the CQT coefficients [25–27]:

$$y^{cq}(k) = \sum_{n=1}^{N_k} [y(n) w_{N_k}(n) \exp(-j2\pi \frac{f_k}{f_s} n)] \tag{2}$$

where $k = 1, \ldots, K$ is the frequency bin index; $w_{N_k}(n)$ is the Hamming window function and N_k denotes variable window lengths; f_s is the sampling rate; and f_k is the center frequency of the bin k. The Q factor is given by

$$Q = \frac{f_k}{f_{k+1} - f_k} \tag{3}$$

To ensure that Q is constant for all frequency bins,

$$N_k = \frac{f_s}{f_k} Q \tag{4}$$

(3) Compute the power spectrum $ps(k)$:

$$ps(k) = |y^{cq}(k)|^2 \tag{5}$$

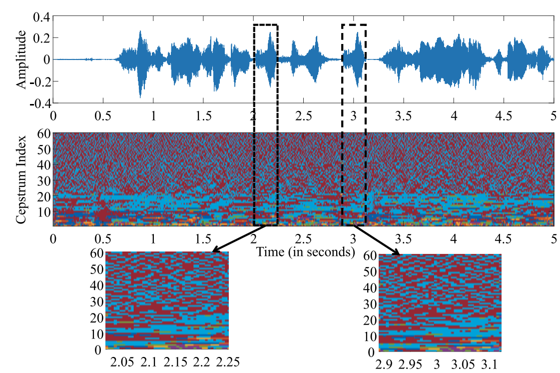

Fig. 3. Illustration of the CQCC feature of an example audio recording.

(4) Separate low and high frequency components in $ps(k)$ by the logarithm:

$$z(k) = \log(ps(k)) \tag{6}$$

(5) Resample $z(k)$ at the uniform sample rate to obtain the linear power spectrum $z(l)$, in which l is the newly resampled frequency bin index [10].

(6) Do DCT on $z(l)$ to extract CQCC:

$$CQCC(p) = \sum_{l=1}^{L} \{z(l) \cos[\frac{p(l-\frac{1}{2})\pi}{L}]\} \tag{7}$$

where $p = 0, 1, \ldots, L-1$, and L is the number of the frequency bins. In this work, the final CQCC feature set is a matrix of 60×36, whose data volume is not high.

It should be noted that although CQCC has been used in speaker verification, the effectiveness of CQCC on the audio CMFD is unknown. Consequently, for a visual understanding of what the CQCC feature can tell about the CMFD, Fig. 3 illustrates the spectral power distribution in different cepstrum index based on CQCC. Note that each color in Fig. 3 represents the number of the same power value in terms of the same cepstrum value. As can be seen from the figure, CQCC can clearly distinguish the similar clips in the forged recording. Consequently, the CQCC feature can be utilized for the CMFD.

4. Similarity computation

In this work, we use PCCs [23] to measure the association between the two pending feature sets. Let t $\mathbf{X} = (x_{ij})_{M\times N}$ and $\mathbf{Y} = (y_{ij})_{M\times N}$ be the CQCC of the two pending short slices. Then, the value of PCCs can be given by

$$r(j) = \frac{\sum_{i=1}^{M}(x_{ij} \times y_{ij}) - \frac{1}{M}\sum_{i=1}^{M} x_{ij} \times \sum_{i=1}^{M} y_{ij}}{\sqrt{\sum_{i=1}^{M}(x_{ij})^2 - \frac{1}{M}(\sum_{i=1}^{M} x_{ij})^2}\sqrt{\sum_{i=1}^{M}(y_{ij})^2 - \frac{1}{M}(\sum_{i=1}^{M} y_{ij})^2}} \tag{8}$$

$$PCC(\mathbf{X}, \mathbf{Y}) = \frac{1}{N}\sum_{j=1}^{N} r(j) \tag{9}$$

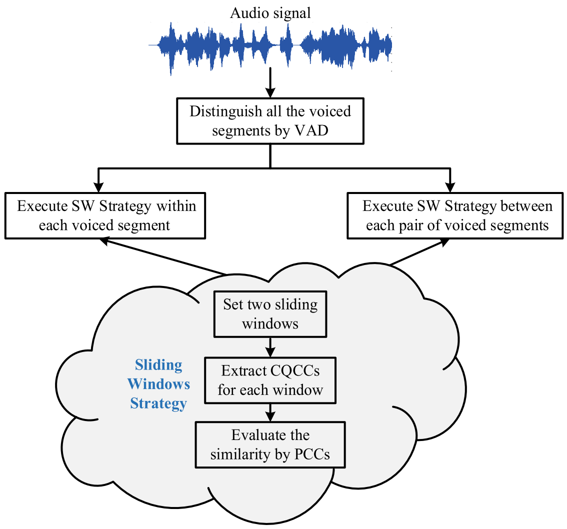

Fig. 4. Framework of the proposed SW-CQCC for the audio CMFD.

where $PCC \in [-1, 1]$. Usually, the closer $|PCC|$ is to 1.0, the more similar $\mathbf{X}$ is to $\mathbf{Y}$.

5. The proposed CMFD method

In this work, we propose a robust CMFD method, named SW-CQCC, as shown in Fig. 4. We first use the spectrum entropy based VAD technique [7,8] to roughly differentiate all the voiced segments. Next, for each voiced segment and each pair of voiced segments, we propose the corresponding SW strategies to divide the voiced segment into overlapping short slices, respectively. Then, we extract the CQCC feature for each window and compute the similarity of the compared two windows with PCCs. Finally, we record the location of the similar windows if the similarity value is beyond the pre-defined threshold. In the following subsections, we will detail each step of SW-CQCC. Particularly, we will try to answer the following question: how do we use the SW strategies to detect and localize short forged clips?

5.1. Partition of voiced segments

In this work, to reduce the impact of background noise, we use a robust spectrum entropy based VAD technique [7,8] to roughly distinguish all the voiced segments. This technique mainly includes the following steps.

(1) The audio signal x is first split into overlapping short-time frames, each of which is applied a window function (e.g., the Hamming window). Denote x_i as the ith frame of x. Do the fast Fourier transform (FFT) on each x_i to obtain the spectral energy $S_i(f)$, in which 250HZ$\leq f \leq$3,500HZ is a frequency component.

(2) Each $S_i(f)$ is decomposed into uniform subbands. Assume that B_j is the set of frequency values in the jth band. The spectral energy of B_j for x_i, $E(j,i)$, is given by

$$E(j,i) = \sum_{f \in B_j} S_i(f) \tag{10}$$

(3) The percentage of each band for spectral energy, $P(j,i)$, and the spectral entropy, $H(i)$, then can be estimated by:

$$P(j,i) = \frac{E(j,i) + \gamma}{\sum_j [E(j,i) + \gamma]} \tag{11}$$

$$H(i) = -\sum_j [P(j,i) \log P(j,i)] \tag{12}$$

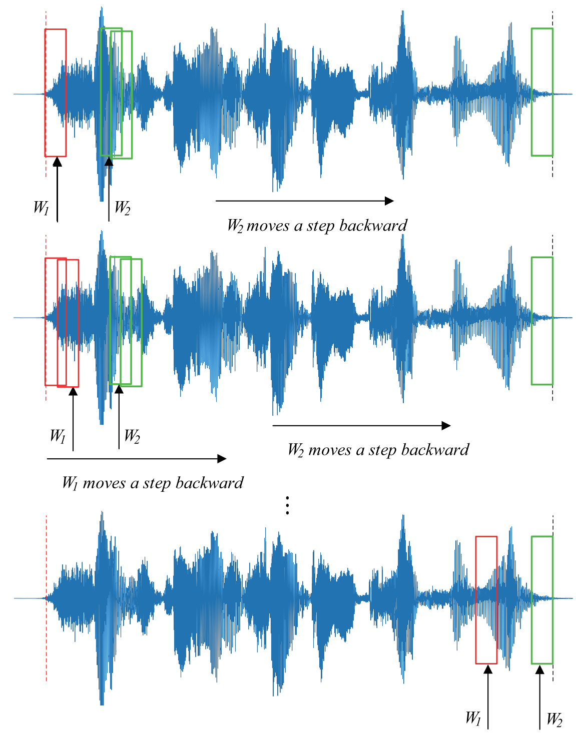

Fig. 5. Illustration of the sliding course in a voiced segment.

where $\gamma > 0$ is a constant for enhancing the discriminability between speech signals and noise signals and improving the robustness of spectral entropy [28].

(4) Make a preliminary check according to a large threshold τ_1:

$$\tau_1 = 0.99 \cdot ave + 0.01 \cdot min \tag{13}$$

where ave and min represent the mean and minimum of $H(i)$, respectively. If $H(i) < \tau_1$, determine the start and end positions of the voiced segment in terms of a small threshold τ_2:

$$\tau_2 = 0.96 \cdot ave + 0.04 \cdot min \tag{14}$$

That is, expand the voiced segment to both sides at the same time until $H(i) < \tau_2$.

It should be noted that the above VAD technique is only used to roughly distinguish all the voiced segments to improve the computational efficiency of SW strategies. Of course any other VAD technique can also be available. To capture the possible short forged slices within and between the distinguished voiced segments, in the following subsections, we will present two SW strategies to further divide voiced segments into short slices for similarity measure.

5.2. SW strategy for each voiced segment

To detect and localize the possible forgeries within a voiced segment, in this subsection, we develop an SW strategy to cut out overlapping short slices within the voiced segment. The basic idea is to set two sliding windows, W_1, W_2, with the same window size t s and the same step size Δs s, as shown in Fig. 5. Notice that in practice, people can speak about 3 words per second, and the pronunciation time of each word is about 0.2s–0.4 s. If a copy-move forgery occurs, there is at least one word between the source and the target slices within the voiced segment. Therefore, an initial interval Δt s exists between W_1 and W_2. Then, the CQCC features of W_1 and W_2 are extracted, respectively, and the similarity between W_1 and W_2 is computed by PCCs. If W_2 is dissimilar to W_1, W_2 moves backward to continue the comparison until W_2 reaches the end. Then, W_1 moves a step backward, W_2 slides again at the position of Δt away from W_1. If the two windows are almost identical, they move backward simultaneously until they are inconsistent. The continuous similar windows just form a copy-move forgery slice.

On the basis of the above idea, the process of the proposed SW strategy for a voiced segment is depicted in Fig. 6, which consists of the following steps.

(1) Initialize W_1 at the beginning of the voiced segment and $W_2 = W_1 + \Delta t$.

(2) Extract the CQCC of W_1 and W_2 and compute the similarity by PCCs.

(3) If the PCC value is smaller than the given threshold Thr, the current W_1 and W_2 are different.

(3.1) If the previous W_1 and W_2 are similar, a forged slice is found, and then let W_1 move forward to the position of the first window of the current forged slice to further determine whether the source has been copied multiple times; otherwise, W_2 moves a step Δs backward.

(3.2) If W_2 is not beyond the end of the voiced segment, Goto (2) to continue the comparison; otherwise, check whether W_1 is a new pending window, namely, whether W_1 has been compared with any other window or not. Specifically, if W_1 has been compared with at least another window, check further whether W_1 belongs to the current forged slice. If no, W_1 does not get a match until W_2 moves to the end of the voiced segment, and thus let W_1 directly move a step Δs backward. If yes, the detection of multiple copies of the current source has finished, and let W_1 move a step Δs backward from the last window of the current forged slice to start the detection of other potential forged slices. Next, if W_1 is a new window, this situation can only occur when W_1 and W_2 move backward together. Then, it is needed to reset W_2 for comparison. In addition, if $W_1 + \Delta t$ is beyond the end of the voiced segment, terminate the similarity comparison and localize all the copy-move forgeries according to the recorded position; otherwise, set $W_2 = W_1 + \Delta t$ and Goto (2).

(4) If $PCC \geq Thr$, W_1 and W_2 are highly similar. Record the position of W_1 and W_2 and then W_1 and W_2 move a step Δs backward at the same time. Goto (3.2) to determine whether the subsequent windows are still similar.

Proposition 1. *If the duration of a pending voiced segment is T s, the number of times that the window slides in Fig. 5 is at least*

$$\frac{T}{\Delta s} + \frac{(T - \Delta t - t)(1 + \frac{T}{\Delta s})}{\Delta s} - \frac{\frac{T}{\Delta s}(\frac{T}{\Delta s} + 1)}{2}$$

Proof. Ideally, there is no forged slice in the pending voiced segment. Then, W_1 slides at least $\frac{T}{\Delta s}$ times. When W_1 slides once, W_2 can only slide behind W_1 and has to slide to the end of the voiced segment. Therefore, the total number of times that W_2 slides is

$$\begin{aligned}
&\frac{T - \Delta t - t}{\Delta s} + \frac{T - \Delta t - t - \Delta s}{\Delta s} + \frac{T - \Delta t - t - 2\Delta s}{\Delta s} \\
&+ \dots + \frac{T - \Delta t - t - \frac{T}{\Delta s}\Delta s}{\Delta s} \\
&= \frac{T - \Delta t - t}{\Delta s}(1 + \frac{T}{\Delta s}) - (1 + 2 + \dots + \frac{T}{\Delta s}) \\
&= \frac{(T - \Delta t - t)(1 + \frac{T}{\Delta s})}{\Delta s} - \frac{\frac{T}{\Delta s}(\frac{T}{\Delta s} + 1)}{2}
\end{aligned}$$

To sum up, the number of times that the window slides is at least

$$\frac{T}{\Delta s} + \frac{(T - \Delta t - t)(1 + \frac{T}{\Delta s})}{\Delta s} - \frac{\frac{T}{\Delta s}(\frac{T}{\Delta s} + 1)}{2} \qquad \square$$

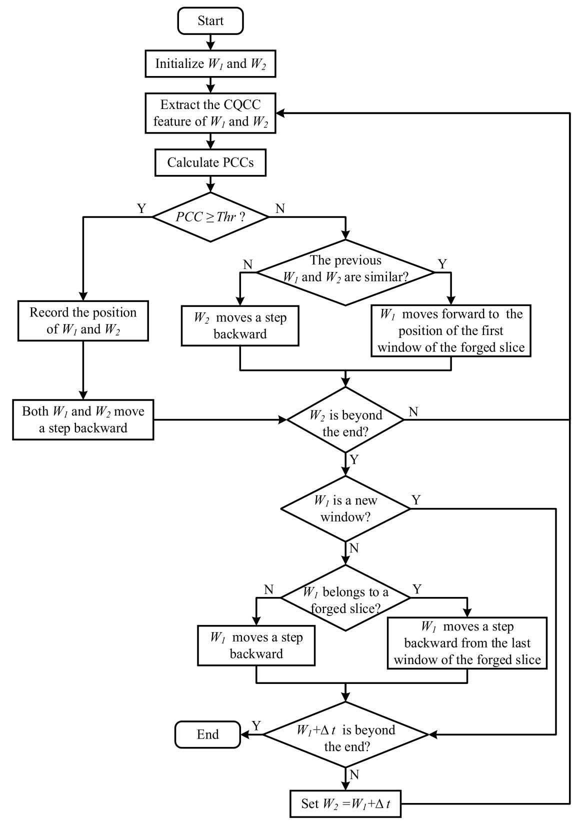

Fig. 6. Basic flowchart of the SW strategy for a voiced segment.

5.3. SW strategy for each pair of voiced segments

To detect and localize forgeries between different voiced segments, in this subsection, we also propose an SW strategy to compare the similarity of windows derived from different voiced segments. As pictured in Fig. 7, the basic idea is similar to Fig. 5. Differently, we set a sliding window W_1 in the first segment (denoted as Segment 1) and the other sliding window W_2 in the second segment (denoted as Segment 2) in terms of the same window size t s and the same step size Δs s. Specifically, both W_1 and W_2 slide from the initial position of their segments. If W_2 is dissimilar to W_1, W_2 continues sliding backward until it reaches to the end of Segment 2. When W_1 moves a step backward in Segment 1, W_2 will slide again from the beginning of Segment 2.

Based on the above considerations, the flowchart of the proposed SW strategy for two different voiced segments is depicted in Fig. 8. The main steps are summarized as below.

(1) Initialize W_1 at the beginning of Segment 1 and W_2 at the beginning of Segment 2.

(2) Extract the CQCC feature of W_1 and W_2, and compute their similarity by PCCs.

(3) If $PCC < Thr$, W_1 and W_2 are dissimilar.

(3.1) If the previous W_1 and W_2 are similar, let W_1 move forward to the position of the first window of the current forged slice in Segment 1 to detect other possible copies in Segment 2; otherwise, W_2 moves a step Δs backward in Segment 2.

(3.2) If W_2 is not beyond the end of Segment 2, Goto (2); otherwise, check whether W_1 is a new pending window in Segment 1. Specifically, if W_1 has been compared with a window in Segment 2, check whether W_1 belongs to the current forged slice in Segment 1. If no, let W_1 directly move a step Δs backward in Segment 1. If yes, let W_1 move a step Δs backward from the last window of the current forged slice in Segment 1 to start the detection of other forged slices. Next, if W_1

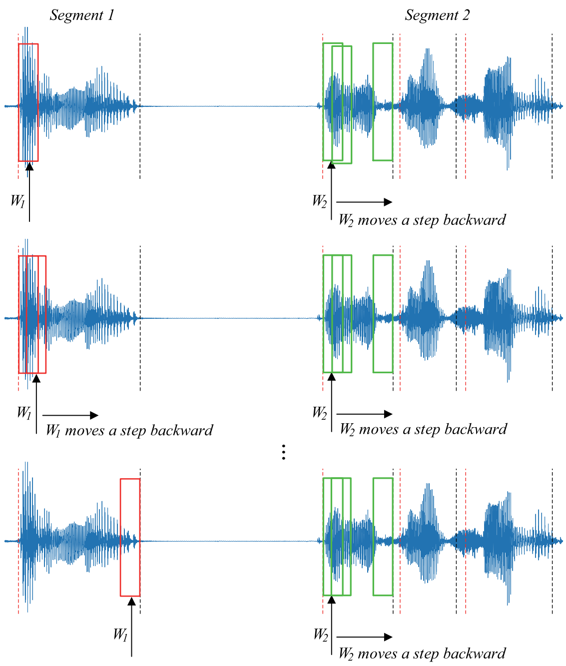

Fig. 7. Illustration of the sliding course between two different voiced segments.

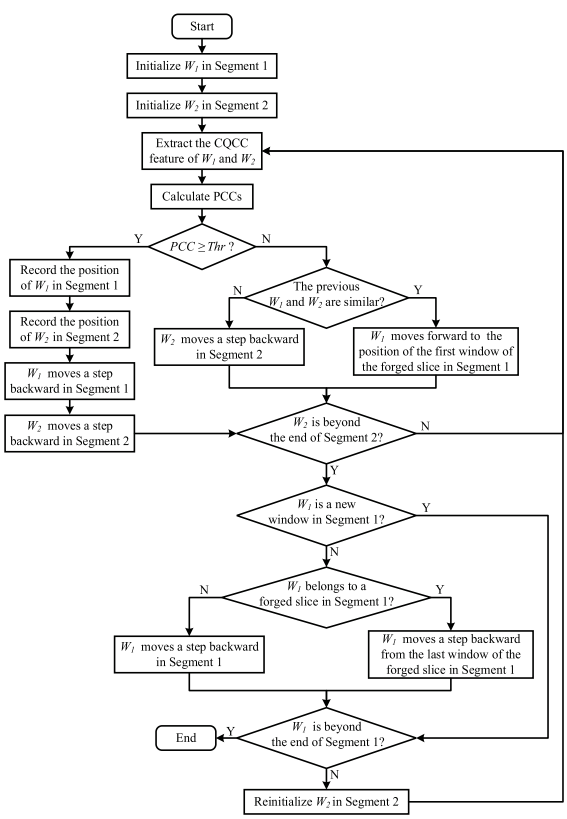

Fig. 8. Basic flowchart of the SW strategy for each pair of voiced segments.

is a new window, it is needed to re-initialize W_2 in Segment 2: if W_1 is beyond the end of Segment 1, output all the copy-move forgeries according to the recorded position; otherwise, re-initialize W_2 at the beginning of Segment 2 and Goto (2).

(4) If $PCC \geq Thr$, record the position of W_1 in Segment 1 and the position of W_2 in Segment 2, respectively. Then, W_1 and W_2 move a step Δs backward at the same time in their respective segment. Goto (3.2) to compare the similarity of the subsequent windows.

Proposition 2. *If the length of the two voiced segments is T_1 s and T_2 s, respectively, the number of times that the window slides in Fig. 7 is at least*

$$\frac{T_1}{\Delta s}(1+\frac{T_2}{\Delta s})$$

Proof. Similarly, assume that there is no forged slice between the two voiced segments. In this case, W_1 slides at least $\frac{T_1}{\Delta s}$ times in Segment 1. For W_2, once W_1 slides to a new position in Segment 1, it has to slide again from the beginning of Segment 2. Thus, the total number of times that W_2 slides is

$$\frac{T_1}{\Delta s}\frac{T_2}{\Delta s}$$

In brief, the number of times that the window slides is at least

$$\frac{T_1}{\Delta s}+\frac{T_1}{\Delta s}\frac{T_2}{\Delta s}=\frac{T_1}{\Delta s}(1+\frac{T_2}{\Delta s}) \quad \square$$

5.4. Discussion of the detection course

The proposed SW-CQCC can be directly used for the audio CMFD. As shown in Fig. 4, for a pending audio recording, we first apply the VAD technique to roughly distinguish all the possible voiced segments. For each voiced segment, we adopt the specifical SW strategy in Fig. 6 to find out all the possible forged short slices within this segment on the basis of CQCC and PCCs. Then, for each pair of different voiced segments, we apply the corresponding SW strategy in Fig. 8 to catch out the forged short slices between the two voiced segments.

The VAD technology can quickly finds voiced segments within large audio recordings. Accordingly, in this work, we use VAD to roughly distinguish the voiced segments to improve the detection efficiency. However, the detection effectiveness of our method does not rely on VAD. This is because we use sliding window strategies to evaluate the similarity between short slices. Particularly, no matter whether the short slice is within a short or a long voiced segment, it will not be missed.

Assume that at total m voiced segments are split out from a pending recording by VAD. Then, the total number of times that the window slides is deduced in Proposition 3. In general, the core idea of the proposed SW-CQCC is to compare and evaluate all the possible short slices within a voiced segment and between different voiced segments at the same time, so as not to miss any likely similar slices.

Proposition 3. *For a pending recording with m voiced segments, the number of times that the window slides is at least*

$$m[\frac{T}{\Delta s}+\frac{(T-\Delta t-t)(1+\frac{T}{\Delta s})}{\Delta s}-\frac{\frac{T}{\Delta s}(\frac{T}{\Delta s}+1)}{2}] + \frac{m(m-1)}{2}\frac{T_1}{\Delta s}(1+\frac{T_2}{\Delta s})$$

Proof. First, each voiced segment needs to be checked, and thus according to Proposition 1, the number of times that the window slides within each voiced segment is

$$m[\frac{T}{\Delta s}+\frac{(T-\Delta t-t)(1+\frac{T}{\Delta s})}{\Delta s}-\frac{\frac{T}{\Delta s}(\frac{T}{\Delta s}+1)}{2}]$$

Second, for m voiced segments, there are total $\frac{m(m-1)}{2}$ pairs of different combinations. Accordingly, based on Proposition 1, the number of times that the window slides between each pair of voiced segments is

$$\frac{m(m-1)}{2}\frac{T_1}{\Delta s}(1+\frac{T_2}{\Delta s})$$

To sum up, the total number of times that the window slides is at least

$$m[\frac{T}{\Delta s}+\frac{(T-\Delta t-t)(1+\frac{T}{\Delta s})}{\Delta s}-\frac{\frac{T}{\Delta s}(\frac{T}{\Delta s}+1)}{2}]$$

$$+\frac{m(m-1)}{2}\frac{T_1}{\Delta s}(1+\frac{T_2}{\Delta s}) \quad \square$$

It should be noted that although the proposed SW-CQCC is composed of several existing mature technologies, such as VAD, CQCC, PCCs, and SW, the effectiveness of the combination of these components in the audio CMFD for blind audio forensics is unknown. In the proposed SW-CQCC, we apply VAD to improve the computational efficiency of the SW strategies due to the fact the silent segments are not worth detection. Additionally, we combine CQCC with PCCs to improve the accuracy of similarity assessment. Moreover, we present SW strategies to improve the robustness of the audio CMFD on short forged slices. In general, the key purpose of this work is to develop a method (i.e., SW-CQCC) to solve the audio CMFD on short forged slices much better than before, which may be beneficial to improving the work efficiency of audio forensic experts.

6. Experimental methodology

In this section, we introduce the compared methods, dataset, parameter values, and performance metrics.

6.1. Compared methods

In this work, we compared SW-CQCC with the state-of-the-art DFT [16,17], LBP [19,20], PFS-DTW [3], and MDCT [22]. As mentioned earlier, DFT [16,17] and LBP [19,20] pay attention to the copy-move forgery on the entire long voiced segment; PFS-DTW [3], MDCT [22], and SW-CQCC address the copy-move forgery on very short slices derived from voiced segments.

The codes of all the compared methods were written in Matlab. All the tests were done on a personal computer with Intel(R) Core(TM) i7-8700 3rd generation CPU @3.20 GHz, 16 GB of DDR4 RAM @ 2666 MHz, and Windows 10 operating system.

6.2. Dataset

We generated two types of copy-move forged datasets based on Librispeech database [29] and a read Chinese speech database (henceforth called Chinspeech). The Librispeech is a popular corpus of read English speech for speech recognition [29]. The Chinspeech is a corpus of read Chinese speech derived from audiobooks on the Internet. The format of the speech is WAV with the sample rate of 16 kHz.

Both the Librispeech and Chinspeech datasets have 500 authentic recordings and 500 forged recordings, each of which is 5 s. For CMFD, the more copy and paste operations are, the easier they are to be detected. Therefore, to fully demonstrate the ability to deal with the forgery of short slices, following the previous practice in [3], we use the CoolEdit audio editor to randomly forge a very short duplicated slice with a duration between 0.2s and 0.4s within a forged recording. For the sake of verifiability, the Librispeech and Chinspeech datasets can be attained at the link: https://github.com/wanna921/Audio-Copy-Move-Forgeries-Datasets.

6.3. Parameter values

The parameters of each method are empirically set as below after a series of experiments. In DFT [16,17], the thresholds of PCCs for the Chinspeech dataset and the LibriSpeech dataset is 0.98 and 0.955, respectively. For LBP [19,20], the threshold of MSE is 50. In PFS-DTW [3], the frame length and the frame shift in YAAPT are 40 ms and 10 ms, respectively; the threshold for DTW distance is 800. In MDCT [22], the frame length and the frame shift in YAAPT are 25 ms and 3 ms, respectively; and the similarity threshold is 0.57. As for SW-CQCC, the frame length and the frame shift in the spectrum entropy based VAD are 25 ms and 10 ms, respectively; $t = 0.19$ s, $\Delta s = 0.01$ s, $\Delta t = 0.48$ s, and $Thr = 0.998$.

6.4. Performance metrics

To comprehensively evaluate the compared methods, in this work, we adopt the popular accuracy, precision, recall, and F-score metrics.

(1) Accuracy is the ratio of both the correctly predicted forged recordings and the correctly predicted authentic recordings to the total recordings.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{15}$$

where TP is the number of forged recordings that are judged as forged recordings; TN is the number of authentic recordings that are judged as authentic recordings; FP is the number of authentic recordings that are judged as forged recordings; FN is the number of forged recordings that are judged as authentic recordings.

(2) Precision is the ratio of correctly predicted forged recordings to the total predicted forged recordings:

$$Precision = \frac{TP}{TP+FP} \tag{16}$$

Obviously, precision focuses on whether an authentic recording is misjudged.

(3) Recall is the ratio of correctly predicted forged recordings to all the forged recordings:

$$Recall = \frac{TP}{TP+FN} \tag{17}$$

which pays attention to whether a forged recording is missed.

(4) F-score is the weighted average of precision and recall:

$$F\text{-}score = \frac{2\times Precision\times Recall}{Precision+Recall} \tag{18}$$

A higher F-score represents better overall performance in terms of precision and recall.

It should be noted that it is difficult to find semantic differences within 0.1 s in an audio recording, and thus the threshold of the absolute error between the actual position of the copy-move forgery and the corresponding localized position is set to 0.1 s. If the absolute error is less than 0.1 s, the pending recording is a forged recording; otherwise, it is an authentic recording.

7. Experimental results and analysis

In this section, we ran experiments and reported the experimental results from different perspectives. Particularly, we will dig deeper into the choice of each key component: why CQCC is selected for the main acoustic features? how to set the window size of the proposed SW strategies? is SW-CQCC available for CMFD? and is SW-CQCC still workable under anti-forensic attacks?

7.1. Comparing different features

To evaluate the effect of the used CQCC features on CMFD, in the first experiment, we compared CQCC with the state-of-the-art speech features for speech recognition: MFCC [24], filterbank (Fbank) [30], spectrogram (Sgram) [31], linear prediction cepstral coefficients (LPCC) [32], and perceptual linear prediction (PLP) [33].

Table 2 depicts the accuracy, precision, recall, and F-score values obtained by our method with different features on different datasets, in which - means the feature failed in detection. The better result between the six features for each dataset is highlighted in boldface. It can be found that the four metric values attained by CQCC on the two datasets are larger than 0.96, while the maximum metric value achieved by MFCC [24], Fbank [30], Sgram [31], LPCC [32], and PLP [33] is smaller than 0.6. It means that our method often failed in detecting the forged slices when it was combined with the latter five features. Conversely, CQCC exhibits high competitiveness and provides a superior performance on CMFD. One explanation is that

Table 2
Experimental results obtained by our method with different features on different datasets.

Feature	Accuracy		Precision		Recall		F-score	
	Librispeech	Chinspeech	Librispeech	Chinspeech	Librispeech	Chinspeech	Librispeech	Chinspeech
MFCC [24]	0.509	0.513	1	1	0.018	0.026	0.035	0.051
Fbank [30]	0.517	0.522	1	1	0.034	0.044	0.066	0.084
Sgram [31]	0.499	0.501	0	1	0	0.002	–	0.004
LPCC [32]	0.508	0.505	1	1	0.004	0.01	0.008	0.02
PLP [33]	0.5	0.5	–	1	0	0	–	–
CQCC	**90.986**	**0.976**	0.998	0.986	**0.974**	**0.966**	**0.986**	**0.976**

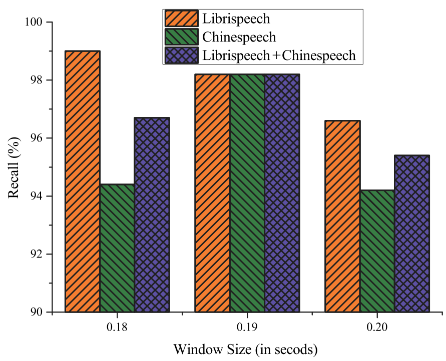

Fig. 9. Recall values obtained by the proposed SW-CQCC under different window sizes.

the compared speech features, such as MFCC [24] and Fbank [30], are commonly used as features in speech recognition systems. They can simulate auditory perception of human's ear well. But when doing speech recognition, the same content of audio clips that one person repeat at different time will definitely be recognized as the same words. This is the reason why these compared speech features are weak in distinguishing the duplicated short slices and the original ones. The other explanation is that in this work, we use PCCs [23] to measure the association between the two pending feature sets, while PCCs [23] in some speech features, such as MFCC [24], have a negative effect on the detection of the short slices [16,21].

7.2. Impact of the window size

The proposed SW-CQCC is based upon the similarity measure of sliding windows. It is natural that the window size may affect the detection performance of SW-CQCC for the forged recordings. To confirm this idea, Fig. 9 shows the recall values obtained by SW-CQCC under different window sizes on the forged recordings of Librispeech and Chinespeech datasets. It can be observed from the figure, when the window size is 0.19 s, SW-CQCC achieved the best overall performance on different datasets. Therefore, in this work, the window size is empirically set to 0.19 s.

7.3. Example of localization

In the third experiment, we use the four example audio recordings in Fig. 1 to investigate the localization performance of the compared methods. Table 3 illustrates the localization results obtained by different CMFD methods, in which Examples 1–4 represent the used audio recordings in Fig. 1(a)–(d), respectively, and - means the method failed in localization. The better result between the four methods for each slice is highlighted in boldface. As can be seen from the table, PFS-DTW [3] failed in localizing a forged slice, DFT [16,17], LBP [19,20], and MDCT [22] made false discoveries and gave wrong location information, while SW-CQCC not only found all the forged slices in the four recordings, but also attained the location of each forged slice with very little error. The above observations indicate that SW-CQCC has better detection and localization capabilities on short forged slices.

7.4. Effectiveness and robustness test

The robustness refers to the ability of the CMFD methods on the dataset with post-processing operations to create anti-forensic attacks. In this experiment, the following five common post-processing operations are chosen to attack the datasets to eliminate the traces of the forgery and secretly create the undetectable forgery. (1) MP3 compression (MPC): MPEG 1 Layer III (MP3) compression at 128 kbps; (2) Gaussian noise addition (GNA-20): Gaussian noise with a signal-to-noise ratio of 20 dB is added; (3) Gaussian noise addition (GNA-30): Gaussian noise with a signal-to-noise ratio of 30 dB is added; (4) Resampling (RS): The recording is first downsampled at 8 kHz, and then upsampled at 16 kHz; (5) Low-pass filtering (LPF): The recording is filtered by the Butterworth filter. These anti-forensic attacks have a great negative impact on the detection effectiveness and will make it more difficult for audio forensic experts to determine whether an audio copy-move forgery occurs.

Tables 4 and 5 show the accuracy, precision, recall, and F-score values obtained by different methods on the two datasets under five anti-forensics attacks, in which - means the method failed in detection under the corresponding attack. The better result between the five methods for each metric under each attack is highlighted in boldface. As can be seen, under various anti-forensics attacks, the overall CMFD performance of all the five methods decreased slightly, but SW-CQCC is significantly better than LBP [19,20], DFT [16,17], MDCT [22], and PFS-DTW [3] for each metric on the two datasets. This stems from the fact that SW-CQCC is superior to LBP [19,20], DFT [16,17], MDCT [22], and PFS-DTW [3] on each metric under each attack, with only one exception. Specifically, under the five anti-forensics attacks, SW-CQCC increased the average values of accuracy, precision, recall, and F-score by about 12.8, 12.6, 7.5, and 13.3 percentage points on the Librispeech dataset, and by about 12.3, 10.6, 10.4, and 12.6 percentage points on the Chinespeech dataset, respectively. Notice that MDCT [22] achieved the worst metric values. An explanation is that in MDCT, the forgery is produced between the voiced segments distinguished by YAAPT, and thus MDCT cannot detect the forged slices within voiced segments. The above observations implies that SW-CQCC is more effective and robust in detecting short forged slices within voiced segments.

7.5. Detection time

Table 6 shows the average detection time (in seconds) of the compared methods on 50 audio recordings with different lengths. As can be seen from the table, for audio recordings with different lengths, our SW-CQCC is much slower than the compared methods, especially on long audio recordings. This reason is that, in order not to miss any suspicious forgery, our SW-CQCC needs to compare and iterate over all the short slices through sliding window strategies. The time consumption is worthwhile in terms of the achieved detection effectiveness.

8. Conclusion and future work

This work investigated the problem of audio copy-move forgery detection (CMFD) in blind audio forensics. We noticed that most of

Table 3
Localization results obtained by different cmfd methods on the example audio recordings.

Example	Actual position		DFT [16,17]		LBP [19,20]		PFS-DTW [3]		MDCT [22]		SW-CQCC	
	Slice 1	Slice 2	Slice 1	Slice 2	Slice 1	Slice 2	Slice 1	Slice 2	Slice 1	Slice 2	Slice 1	Slice 2
1	1.42s ∼ 1.69s	3.00s ∼ 3.27s	0.57s ∼ 0.80s	1.57s ∼ 1.80s	–	–	–	–	–	–	**1.41s ∼ 1.68s**	**2.99s ∼ 3.26s**
2	3.26s ∼ 3.49s	4.52s ∼ 4.75s	–	–	2.09s ∼ 2.1s	4.7s ∼ 4.8s	–	–	–	–	**3.31s ∼ 3.54s**	**4.58s ∼ 4.81s**
3	1.79s ∼ 2.00s	4.12s ∼ 4.33s	–	–	0.86s ∼ 1.02s	1.09s ∼ 1.25s	–	–	2.32s ∼ 2.73s	4.42s ∼ 4.45s	**1.79s ∼ 1.99s**	**4.11s ∼ 4.31s**
4	0.55s ∼ 0.73s	1.66s ∼ 1.84s	–	–	–	–	–	–	–	–	**0.53s ∼ 0.72s**	**1.65s ∼ 1.84s**

Table 4
Metric values attained by different methods under five anti-forensics attacks on the Librispeech dataset.

Attack	Accuracy					Precision				
	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC
No attack	67.7	76.8	86.2	50.8	**98.6**	64.68	80.04	89.7	51.4	**99.8**
MPC	70.6	78.8	84.6	51.3	**95.2**	67.58	78.8	87.1	52.06	**96.5**
GNA-20	50.2	50	78.9	48	**92.8**	50.1	–	83.5	46.53	**96.3**
GNA-30	52.4	50.8	82	50.3	**95.7**	51.48	83.33	85.9	50.47	**97.1**
RS	67.2	78.1	80.3	49.9	**93.6**	63.56	78.27	81.2	49.85	**96.4**
LPF	67	77.8	80.5	50.7	**92.8**	63.32	77.58	79.8	51.02	**94**
Average	61.48	67.1	81.26	50.04	**94.02**	59.21	79.6	83.5	49.99	**96.06**
Attack	Recall					F-score				
	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC
No attack	78	71.4	81.8	29.4	**97.4**	66.15	78.39	85.6	51.1	**98.6**
MPC	79.2	78.8	81.2	32.8	**93.8**	69.06	78.8	84	51.68	**95.1**
GNA-20	**97.2**	0	72	26.8	89	50.15	–	77.2	47.25	**92.5**
GNA-30	83.6	2	76.6	32	**94.2**	51.93	63.12	81	50.39	**95.6**
RS	80.6	77.8	78.8	33.6	**90.6**	65.33	78.18	80	49.88	**93.4**
LPF	80.8	78.2	81.6	35	**91.4**	65.11	77.69	80.7	50.86	**92.7**
Average	84.28	47.36	78.04	32.04	**91.8**	60.32	75.24	80.58	50.01	**93.86**

Table 5
Metric values attained by different methods under five anti-forensics attacks on the Chinespeech dataset.

Attack	Accuracy					Precision				
	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC
No attack	74.5	72.3	86.8	53.7	**97.6**	73.06	83.89	91.3	55.73	**98.6**
MPC	76.2	74.8	85.4	52.3	**96.7**	75	81.31	88.6	53.56	**97.9**
GNA-20	50.8	50	79.1	53.5	**92.2**	50.4	–	84.2	56.18	**94.1**
GNA-30	54.3	50	82.3	53	**95.5**	52.96	–	89.4	54.55	**96.7**
RS	75	73.1	80.8	54.2	**94.5**	73.23	82.91	80.6	56.4	**97.6**
LPF	74.7	72.5	80.5	54	**90.5**	73.08	80.49	81.2	55.95	**90.9**
Average	66.2	64.08	81.62	53.4	**93.88**	64.94	82.15	84.8	55.33	**95.44**
Attack	Recall					F-score				
	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC
No attack	77.6	55.2	81.4	36	**96.6**	73.78	77.67	86.1	54.69	**97.6**
MPC	78.6	64.4	81.2	34.6	**95.4**	75.595	77.92	84.7	52.92	**96.6**
GNA-20	96.4	0	71.6	31.8	**90**	50.61	–	77.4	54.81	**92**
GNA-30	77	0	76	36	**94.2**	53.62	–	82.2	53.76	**95.4**
RS	78.8	58.2	81.2	37	**91.2**	74.11	77.69	80.9	55.28	**94.3**
LPF	78.2	59.4	79.4	37.6	**90**	73.88	76.29	80.3	54.96	**90.4**
Average	81.8	36.4	77.88	35.4	**92.16**	65.56	77.39	81.1	54.35	**93.74**

Table 6
Average detection time (in seconds) for 50 audio recording.

Length (s)	LBP [19,20]	DFT [16,17]	PFS-DTW [3]	MDCT [22]	SW-CQCC
2 s	0.239	0.072	0.341	0.384	4.774
5 s	0.734	0.082	0.963	0.977	7.432
10 s	1.24	0.162	1.784	1.717	21.287

the existing methods are good at detecting the forgery of the entire, long voiced segment, but ignore the possible forgeries of short slices derived from voiced segments, especially when the source and the target slices are within the same voiced segment. To this end, we used the constant Q cepstral coefficients (CQCC) as the main acoustic feature and the Pearson correlation coefficients to measure the association between two feature sets. Then, we presented a robust audio CMFD method, called SW-CQCC, on the basis of rough pre-segmentation and two sliding window (SW) strategies, which are intended for similarity evaluation within each voiced segment and between each pair of voiced segments, respectively. Finally, we evaluated SW-CQCC through extensive experiments against the state-of-the-art methods to CMFD on real-world copy-move datasets with read English and Chinese corpus.

Experimental results demonstrate that SW-CQCC exhibits a superior performance in CMFD and is more effective and robust against various anti-forensics attacks, which may be beneficial to the audio forensic expert in audio evidence collection.

It should be noted that our SW-CQCC has the following limitations. On one hand, the sliding window strategies in our SW-CQCC may be time-consuming for long audio recordings. On the other hand, similar to the existing works, our SW-CQCC also requires a large amount of experimental effort to select the appropriate similarity threshold for the used dataset. Consequently, how to improve the detection efficiency and achieve the adaptive threshold selection for similarity evaluation on different datasets will be studied in depth in our future work.

CRediT authorship contribution statement

Zhaopin Su: Conceptualization, Methodology, Writing. **Mengke Li:** Software, Data curation. **Guofu Zhang:** Writing, Editing. **Qinfang Wu:** Software, Visualization. **Yaofei Wang:** Reviewing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Datasets can be attained at the link: https://github.com/WuQinfang/CMFD/

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